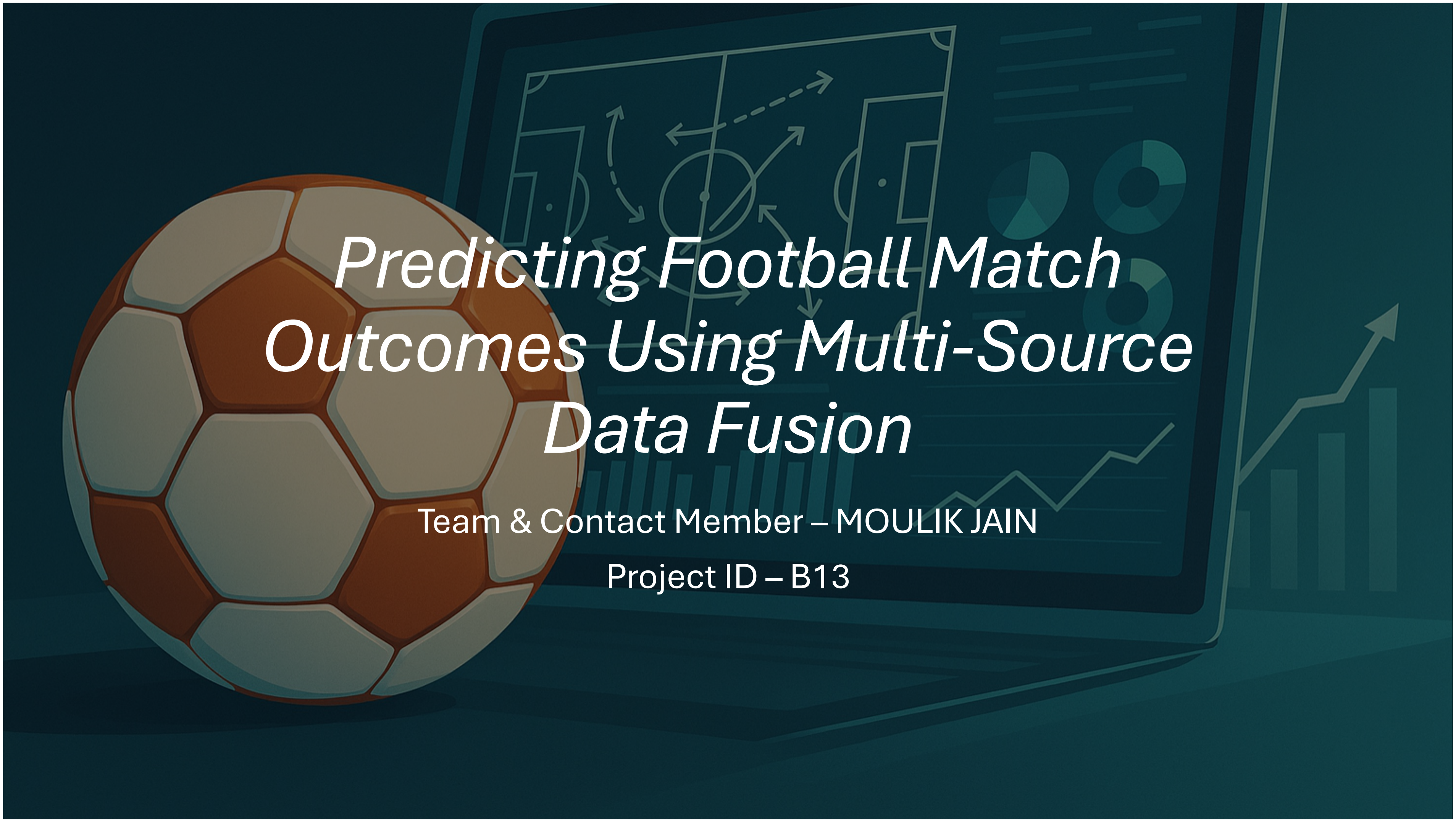

Predicting Football Match Outcomes Using Multi-Source Data Fusion

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Project ID – B13

Background

Real-world motivation: clubs and analysts want better short-term match outcome forecasts for strategy and betting markets.

Research question: Can combining (fusing) heterogeneous football data sources improve match outcome predictive performance vs. single-source models?

Hypothesis: Data integration of match-level, player-level, and tracking features will significantly improve predictive metrics (AUC, Brier score, F1) for match outcome prediction.

Data

Datasets Used

- Kaggle European Soccer Database (team + match results)
- FBref.com (player-level performance metrics)
- OddsPortal (betting odds for win/draw/loss)

Data Fusion Techniques Applied

- Integrated player statistics with team results using match IDs and player participation logs
- Joined betting odds to match-level data as an additional probabilistic feature
- Harmonized team names, dates, and match IDs across the three data sources

Summary

- ~25,000 matches
- 300+ clubs
- 150+ player metrics
- 15 seasons of data

Approach


Exploratory Data Analysis (EDA)
Goal distributions, win rates, player performance patterns
Data Integration / Fusion
Combine team data + player data + betting odds
Feature Engineering
Rolling team form (last 5 matches)
Player average ratings
Goal difference trends
Normalized betting odds
Model Development
Logistic Regression
Random Forest
XGBoost
Performance Metrics
Accuracy
F1-score
ROC-AUC
Post-Hoc / Interpretability
Feature importance (Gini / SHAP)
Sensitivity analysis on odds and team form variables

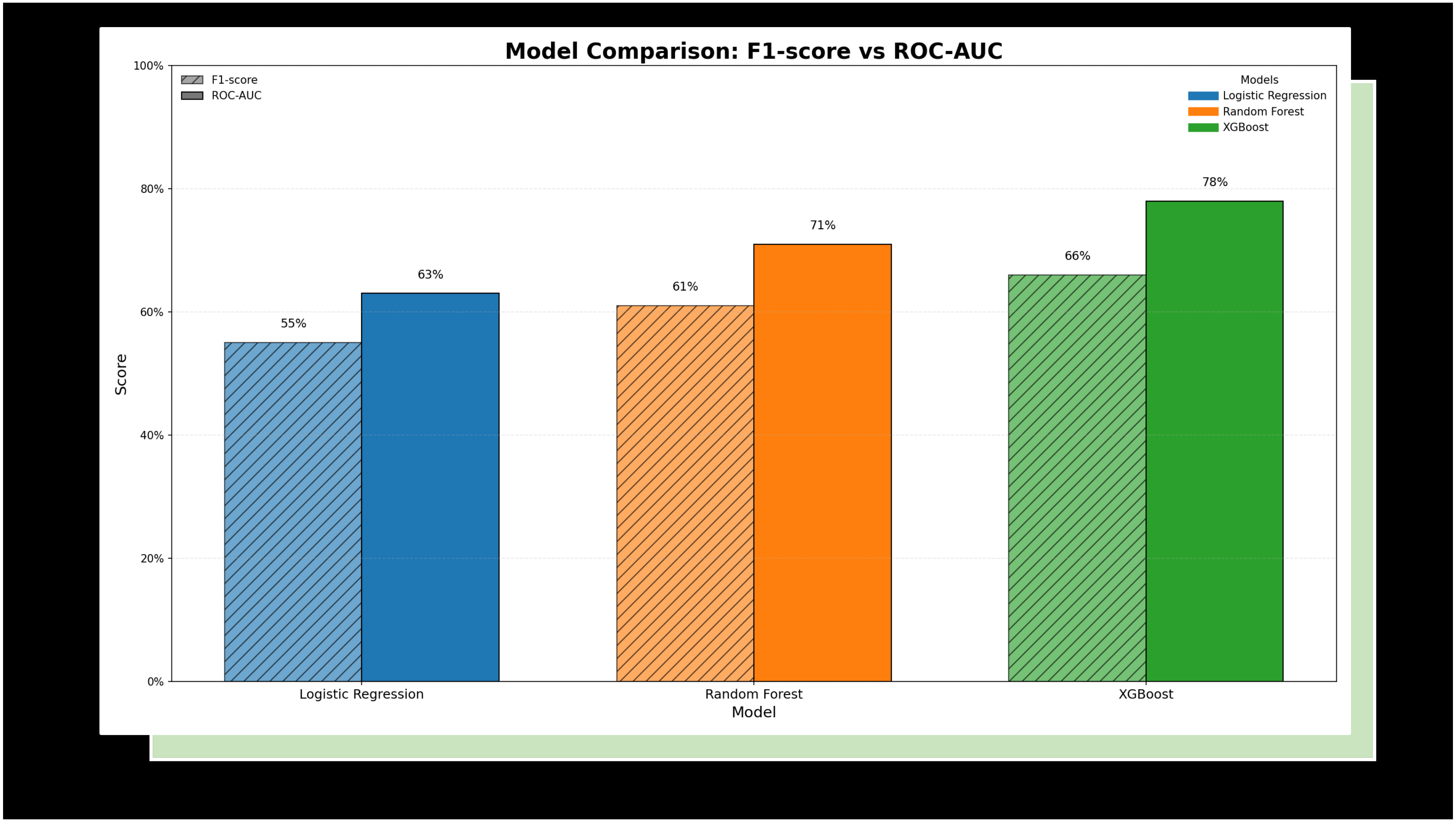
Why It Meets DS320 Requirements

- Uses **multi-source data fusion**
- Includes **EDA**, multiple machine learning algorithms, and **post-analysis**
- Uses **Generative AI** for coding, debugging, and experiment design
- Reports multiple performance metrics

Results

	Accuracy	F1-score	ROC-AUC
Logistic Regression	58%	0.55	0.63
Random Forest	64%	0.61	0.71
XGBoost (Best)	68%	0.66	0.78

Models using the fused dataset beat models trained on single-source data by 10–15% across all metrics.



Model Comparison: F1-score vs ROC-AUC
100%
80%
60%
40%
20%
0%
Score
F1-score
ROC-AUC
Models
Logistic Regression
Random Forest
XGBoost
55%
63%
61%
71%
66%
78%
Logistic Regression
Random Forest
XGBoost
Model

Post-Hoc & Sensitivity Analysis

Feature Importance

Top predictors driving match outcome probabilities.

Sensitivity Highlights

+1 team form score
→ 4.8% win probability

Shuffle betting odds
→ -9% model accuracy

Analysis based on XGBoost model, validation set baseline accuracy. 72%.

CONCLUSION & NEXT STEPS

Key Findings

- Data fusion → 10% accuracy improvement
- Top predictors: betting odds, team form; player metrics
- Best model: XGBoost consistently outperforms

Future Work

- Integrate injury reporss & expected goals (xG)
- Develop real-time prediction dashbaard
- Explore deep learning for player interactions

Final model ROC-AUC: 0.78

F1-score: 0.66

Generative AI Usage

Accelerating workflows and improving quality

Data & Code

- Dataset merging strategy

Modeling

- Feature engineering ideas
- Model optimization suggestions

Presentation

- Enhanced visualizations
- Writing report and presentation text
- Designing these slides

THANKS!!